H2-A-01 result report: one recorded RHF/STO-3G point

Status. Unpublished research note: an [AGENT] summary, prepared 2026-10-29, of one recorded software output and of private task records. Method and scope: RHF/STO-3G at a fixed 1.4 bohr separation, one run, one point. It authorizes nothing and makes no claim about any other calculation. Record: results/h2/H2-A-01-run-01.

Derived summary of the full report docs/learning/h2/H2-A-01-RESULT-REPORT.md, SHA256 9289a4c30ebaeed69b1c318fbd296eb2be6d3e4289c051502f3620940c08ed35, listed in provenance/EXPORT-MANIFEST.md. It reproduces marked blocks of the report in the report's own wording; no statement appears only here. Tables and detail are in the full report; section numbers are those of the full report.

1. Summary and scope

One bounded attempt of job H2-A-01 ran once on 2026-10-29 through the route-R2 driver and ended with the driver tier COMPLETED_PENDING_REVIEW. The engine (Psi4 1.11) reported a total energy of **-1.11671432506257 hartree** for the hydrogen molecule at one fixed, input separation of **1.4 bohr**: RHF/STO-3G, two electrons, singlet, converged in 2 SCF iterations (8.9037 s, exit 0). The 500 MiB memory figure was the engine's memory setting, not an operating-system memory cap. Locators, in the results record: raw/result.json; redacted/psi4-output.dat lines 153 to 155, 177 and 184; redacted/driver-outcome.json. The values are recorded engine output with [COMP-PRED] not assigned (section 7); the settings are [AGENT] proposals as applied.

What this is: a single-point RHF/STO-3G number at one fixed separation, recorded from one run and narrowly reviewed.

What this is not: a dissociation curve; an optimized or predicted equilibrium bond length (1.4 bohr is the fixed input internuclear distance, not a result); readiness for any further work; parity with any other calculation; a reproduction of, or agreement with, the benchmark study; or validation of any donor, surface, pathway or diagnosis of the stopped E-01 attempt.

Residuals that this report states and does not resolve (sections 7 to 9): the engine's source-code identity is unestablished and no [COMP-PRED] is granted; the CURRENT ENERGY alias is a [GAP]; OMP Warning #179 is INDETERMINATE with its harmlessness unproved; and the Lead's disposition (section 7) records the reviewed-calculation requirement as not met.

Figure 3. Fixed input geometry

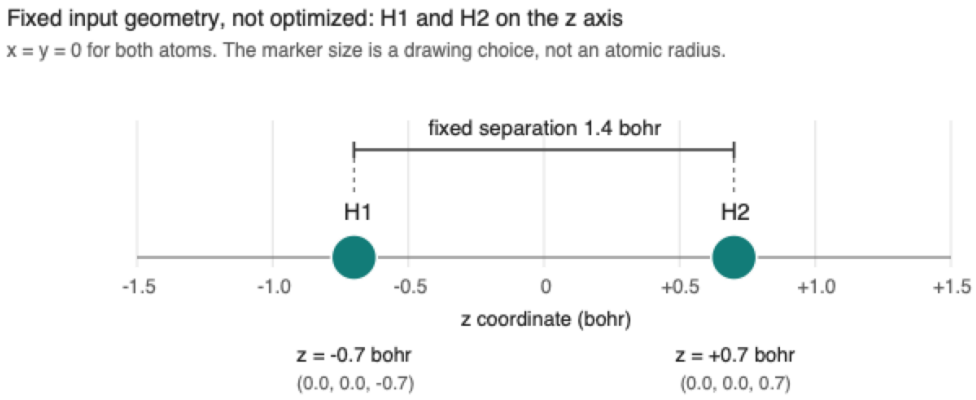


Figure 3. Fixed input geometry of H2-A-01: H1 and H2 on the z axis at -0.7 and +0.7 bohr. The 1.4 bohr internuclear distance is the fixed input, not an optimized or predicted equilibrium bond length, and not a result. Source table and renderer: section 5 of the full report.

4. What the reported total includes

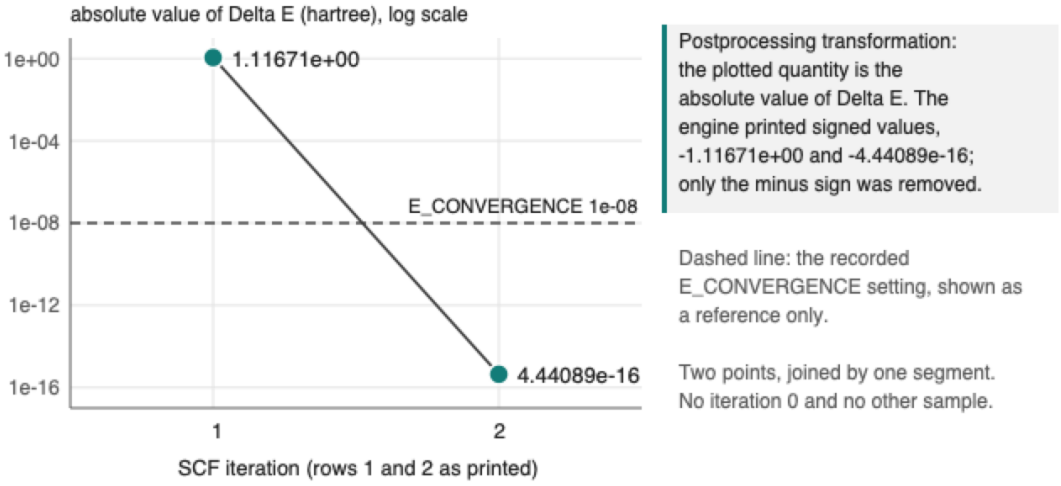
The total energy that the engine returned **includes the nuclear repulsion** for this build. The independent review established this from the engine's printed decomposition: the nuclear repulsion, the one-electron energy and the two-electron energy sum to the printed total to 1e-16 hartree (`psi4-output.dat` lines 181 to 184). Under that convention, attributed to the independent review, the electronic energy is the reported total minus the reported nuclear repulsion, -1.8310000393482843 hartree; it **includes the electron kinetic energy** (the lesson's convention, in its section "How the Hamiltonian splits") and the engine did not print it.

Figure 4. SCF convergence, as printed

SCF convergence of job H2-A-01: the two printed iterations

Source: the engine's printed iteration table (psi4-output.dat lines 151, 153 and 154). Nothing else was sampled.

A. Absolute value of the printed Delta E, logarithmic axis



B. The two printed rows, verbatim (signed Delta E)

		Total Energy	Delta E	RMS |[F,P]|	
@RHF iter	1:	−1.11671432506257	−1.11671e+00	1.56620e−16	DIIS
		=		=	
@RHF iter	2:	−1.11671432506257	−4.44089e−16	1.56620e−16	DIIS

An = between the rows marks a column whose two printed strings are identical.

Delta E is quoted as printed; it is not recomputed from the rounded totals. The engine printed no iteration 0 row.

Units: Total Energy and Delta E are energies in hartree (Eh).

`RMS |[F,P]|` (third column): the engine prints no unit.

Figure 4. Panel A: the absolute value of the engine's printed, signed Delta E at SCF iterations 1 and 2, on a logarithmic axis; taking the absolute value is a postprocessing transformation (only the minus sign was removed). The dashed line is the recorded E_CONVERGENCE setting, a reference only. Panel B: the engine's own two rows, verbatim, with no iteration 0 and no other point; the third column is the engine's `RMS |[F,P]|`, not a density change. Source table and renderer: section 5 of the full report.

Figure 5. Signed energy components

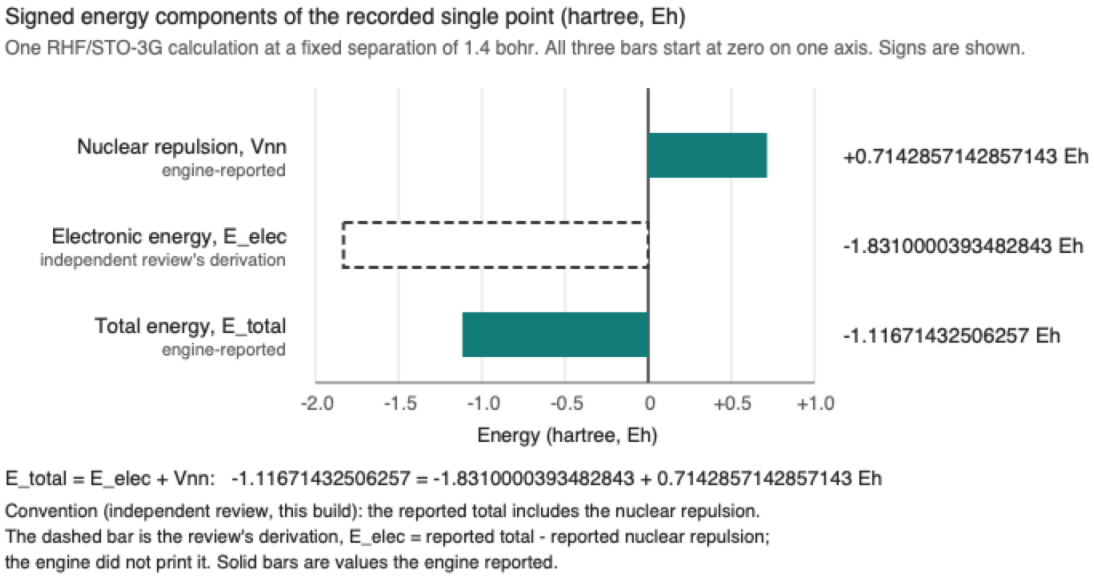


Figure 5. Signed energy components of the recorded single point, in hartree, all anchored at zero on one axis, with E_total = E_elec + Vnn. Vnn and E_total were reported by the engine (solid bars). E_elec is the independent review's derivation under the convention that the reported total includes the nuclear repulsion (dashed outline): the engine did not print it. Source table and renderer: section 5 of the full report.

7. Review, chronology and classification

Chronology, 2026-10-29, nothing rewritten: an independent review of the run result recorded APPROVE, and the Lead accepted the narrow result; a review of the material behind a later host-capture summary recorded **REJECT** with one blocker (two checks the retained capture does not contain), which stays on the record; a documentary erratum relabelled those claims supplied and unverified, and a further review recorded APPROVE; a final review recorded APPROVE, and the Lead's disposition records the orchestration run-record requirement as **met on a retained supplied-source basis only** (its external files and store digests were supplied, not independently authenticated) and the reviewed-calculation requirement as **not met**, because the engine's source-code identity is unestablished. The step-by-step chronology is in the full report.

Classification. The values are actual software output of one engine run. `[COMP-PRED]` is **not granted**: the evidence policy (section 4) requires complete provenance, and the remaining gap is the engine's source-code identity (section 8). The classification belongs to the independent review and Lead acceptance, recorded privately.

8. Engine source-code identity: the remaining evidence-policy gap

The engine is the conda-forge psi4 1.11 build `py314h53d0584_1` (archive SHA256 `92d4aeb73fe7027bd239a656cf353461490420d76096ae762bd9f1c29ed4da10`). The code-provenance record excludes the engine environment; the engine prints the build's placeholder revision, `Git: Rev {} zzzzzzz` (`redacted/psi4-output.dat` line 6). **Source-to-binary correspondence is not established** and no `[COMP-PRED]` is granted; the catalogue (docs/SOURCES.md [31], [37], [52], [53], [54], quoted in the full report) records that the material this project holds cannot establish it.

Two anomalies are explained by **supplied** evidence, a read-only observation of the external engine environment made outside this project's boundary: the installed core extension differs from its archive path record because conda applied its declared prefix replacement and re-signed it ad hoc, matching conda's own install-time record; and the banner revision is the shipped version metadata's placeholder. This shows no tampering or different build, an absence of evidence and not a proved negative; the patch and re-signing were not reproduced byte for byte.

The run's own identity record lists 468 loaded images, including `libint2.dylib` at the engine prefix, a runtime observation; a path is not a build identity and the load name encodes no version (compatibility version 0.0.0 in the supplied static check), so the loaded build, version and source linkage stay unresolved.

9. Open items and residuals

- **The returned-energy alias is a GAP.** For `CURRENT ENERGY`, the payload's source text is a literal string at line 445 of the payload at the base commit, not a read-back. The returned, printed-final and printed-total energies agree (`psi4-output.dat` lines 177 and 184; `psi4-output.log` line 3), but the label is not independently proved.
- **OMP Warning #179, INDETERMINATE.** `raw/engine.stderr` line 1 reports a file-sizing failure; its cause and numerical impact are unknown, its harmlessness is unproved, and convergence does not prove it harmless.
- **The Lead's disposition (section 7).** The result has not been ingested into the orchestration system (the approved text does not require it) and this report does not claim it. The reviewed-calculation requirement is **not met**: its provenance clause is open on the engine's source-code identity (section 8).
- Also open: the `network.prerequisite` FAIL; host provenance (section 6); P1, loaded identity, the Libint load-name question, E1, profile compatibility; M1; retention and execution completeness; Phase B NOT_RUN; E-01 unexplained.

10. Research connection and a conceptual question

Why a two-electron calculation is in this repository. The research aims at a reviewed, reproducible computational capability for mechanically controlled chemistry, starting with engine and method validation on well-defined species before any tool-surface model is attempted (.agents/RESEARCH-STATE.md, section 1; README.md, opening paragraph). The chemistry in view is the benchmark's carbon donation from a germanium-based tool to a hydrogen-passivated silicon surface (docs/benchmark.md, "Experimental setup"). The hydrogen molecule is the smallest case showing every step of such a calculation (docs/learning/h2/README.md, opening paragraph).

What the calculation does. The nuclei are held fixed, a computational choice and not a claim that they are classical, so the electrons form a problem that depends on the positions only as a parameter (lesson, "Why the nuclei are held fixed", citing SOURCES [45]). The program improves the orbitals by repetition (lesson, "What SCF varies", citing [48]); "converged" means only that the reported criteria fell below their thresholds, not that the result is a minimum or the best answer (lesson, "Convergence is not a minimum"); here two iterations sufficed (Figure 4). RHF is a single determinant and STO-3G gives two basis functions, so the number depends on both and on the distance chosen as input (lesson, "What RHF and a finite basis leave out" and "Single point, scan, and optimization"). The total includes the nuclear repulsion and the electronic term includes the electron kinetic energy (section 4).

The shared bookkeeping. The run exercised, at tiny scale, the provenance that the evidence policy asks of any result (docs/evidence-policy.md, section 4), and showed where it falls short (section 8).

What this is not. It is a tiny two-electron learning calculation. There is **no converged donor result**, and the coupled tool-surface route is **not demonstrated** (AGENTS.md, "Claims that require reviewed evidence", bullet "Coupling"). E-01, the attempt aimed at the donors, did run a precursor and stopped during integral setup; it is closed as technically blocked and scientifically indeterminate, its cause was never diagnosed, and it is retained (results/e01/README.md, opening paragraph). The donor and surface models are unoptimized definitions (RESEARCH-STATE.md, section 3), and the donor, surface and coupling work remains unvalidated.

A conceptual question. This number depends on a basis set, an electronic model, a fixed distance and an engine whose source identity is not established. For a calculation on the donor, which of those dependencies would a reader most need to be able to check from the record alone, and what would that check consist of?